

AI Lab 4

1. A learner stores the training examples given below in a table and answers a query by looking it up. Write a program to build the table and answer the four queries. State what the learner can do for a query it has never seen.

```
Marks : 45  52  61  73  88
Grade :  D   C   C   B   A

Queries : 61, 45, 65, 90
```

(Learning by Memorization)

2. For the training data given below, classify the test point using the **k-Nearest Neighbour** method with Euclidean distance. Print the distance from the test point to every training point, and the predicted class for **k = 1** and **k = 3**.

```
Point   x1  x2  Class      Test point : (3, 4)
P1      1   2   A
P2      2   3   A
P3      3   3   B
P4      6   5   B
P5      7   7   B
```

(k-Nearest Neighbour)

3. For the training data given below, use the **Naive Bayes** classifier to decide whether Play is Yes or No for the new day. Print the prior probability of each class, the conditional probability of each attribute value, and the final score of both classes.

No.	Outlook	Temperature	Play
1	Sunny	Hot	No
2	Sunny	Hot	No
3	Overcast	Hot	Yes
4	Rainy	Mild	Yes
5	Rainy	Cool	Yes
6	Rainy	Cool	No
7	Overcast	Cool	Yes
8	Sunny	Mild	No
9	Sunny	Cool	Yes
10	Rainy	Mild	Yes

New day : Outlook = Sunny, Temperature = Cool

(Naive Bayes)